

4. Polyalphabetic Substitution Cipher

```
#include <stdio.h>

#include <string.h>


int main()
{
    char text[100], key[100], cipher[100];
    int i, j, len, klen;


    printf("Enter plaintext: ");
    scanf("%s", text);


    printf("Enter key: ");
    scanf("%s", key);


    len = strlen(text);
    klen = strlen(key);


    for(i = 0, j = 0; i < len; i++, j++)
    {
        if(j == klen)
            j = 0;


        cipher[i] = ((text[i] - 'A') +
                     (key[j] - 'A')) % 26 + 'A';
    }
```

```
cipher[len] = '\0';
```

```
printf("Ciphertext: %s", cipher);
```

```
return 0;
```

```
}
```

